



THE DATA SCIENCE COLLABORATORY AT COLGATE UNIVERSITY

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ABOUT THE COLLABORATORY

- The Data Science Collaboratory (DSC) is a data analysis collaboration network focused exclusively on smaller colleges and universities in New York State.
- Leveraging expertise across the New York Six Liberal Arts Consortium (NY6), the DSC consists of a cohort of statisticians and data librarians from regional colleges and universities and from within the community.
- Student collaborators of the DSC contribute to the development of R Shiny [1] applications that provide data visualization tools and statistical procedures.
- As of Fall 2024, students and faculty can now get statistical consultation at the new DSC desk on the fifth floor of Case-Geyer Library.

FOCUSES FOR THE SUMMER

- Improve the user experience of the R Shiny applications in regards to responsiveness and feedback, as well as making statistical interpretations clearer and comprehensive.
- Version 2.0 apps have enhanced workflows and computational efficiency.
- Revamp the Resampling App.
- Develop new apps for Mixed Effects Regression, Robust Regression, and Two-Way ANOVA, to expand the types of questions researchers can answer using DSC apps.

ABOUT THE DESK

- The DSC desk is there so that students working with data or a new coding language can stop by and ask the collaborators any questions they have.
- We can help with:
 - R, Python, MATLAB, and Excel
 - Cleaning data
 - Making graphs via the `tidyverse` [4] and `ggplot2` [3] packages
 - Interpreting/Analyzing results
 - Questions about statistical concepts

FUTURE GOALS

- We are working to create/finish the following:
 - Wilcoxon and Mann-Whitney Test
 - Power Analysis
 - Ordinal Regression
 - Poisson Regression
 - Negative Binomial Regression
- We will also continue to enhance the user interface to maximize usability and efficiency
- Larger presence on campus by increasing the outreach for the DSC desk.

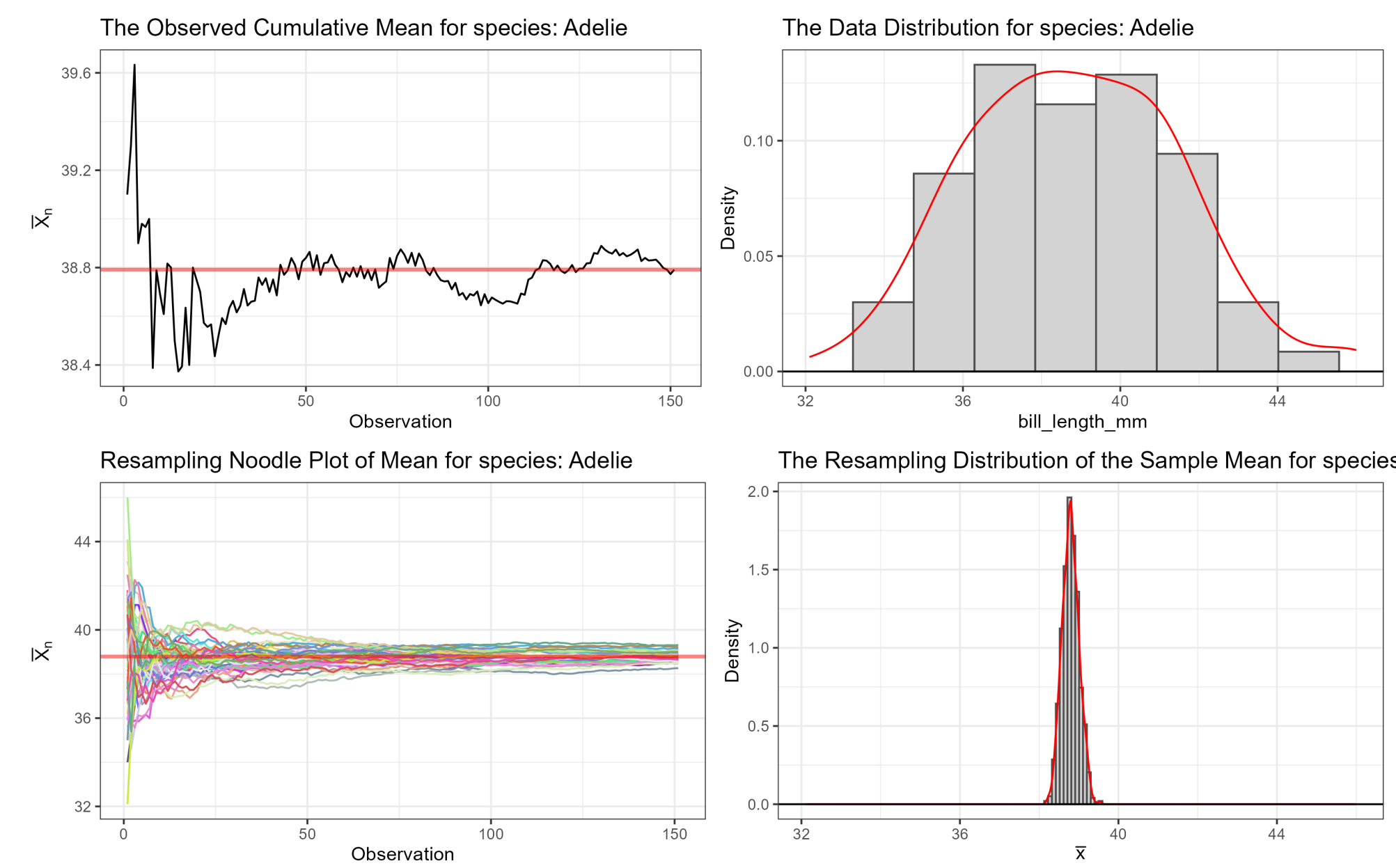
LINK TO APPS

<https://shiny.colgate.edu/apps.html>

REFERENCES

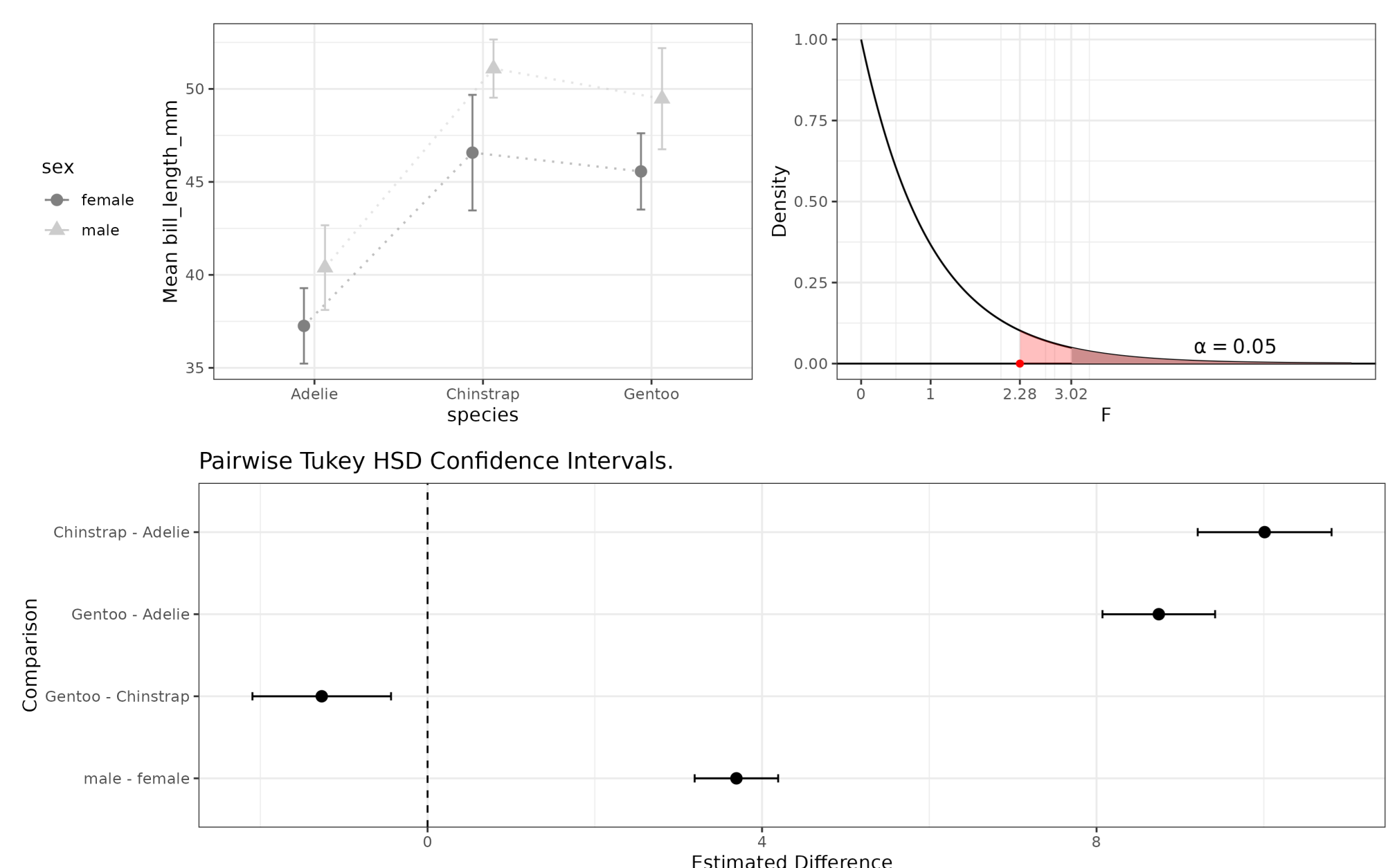
- [1] Winston Chang, Joe Cheng, JJ Allaire, Carson Sievert, Barret Schloerke, Yihui Xie, Jeff Allen, Jonathan McPherson, Alan Dipert, and Barbara Borges. *shiny: Web Application Framework for R*, 2021. R package version 1.6.0.
- [2] Allison Marie Horst, Alison Presmanes Hill, and Kristen B Gorman. *palmerpenguins: Palmer Archipelago (Antarctica) penguin data*, 2020. R package version 0.1.0.
- [3] Hadley Wickham. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York, 2016.
- [4] Hadley Wickham, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemond, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Müller, Jeroen Ooms, David Robinson, Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani. Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686, 2019.

RESAMPLING ADDITIONS



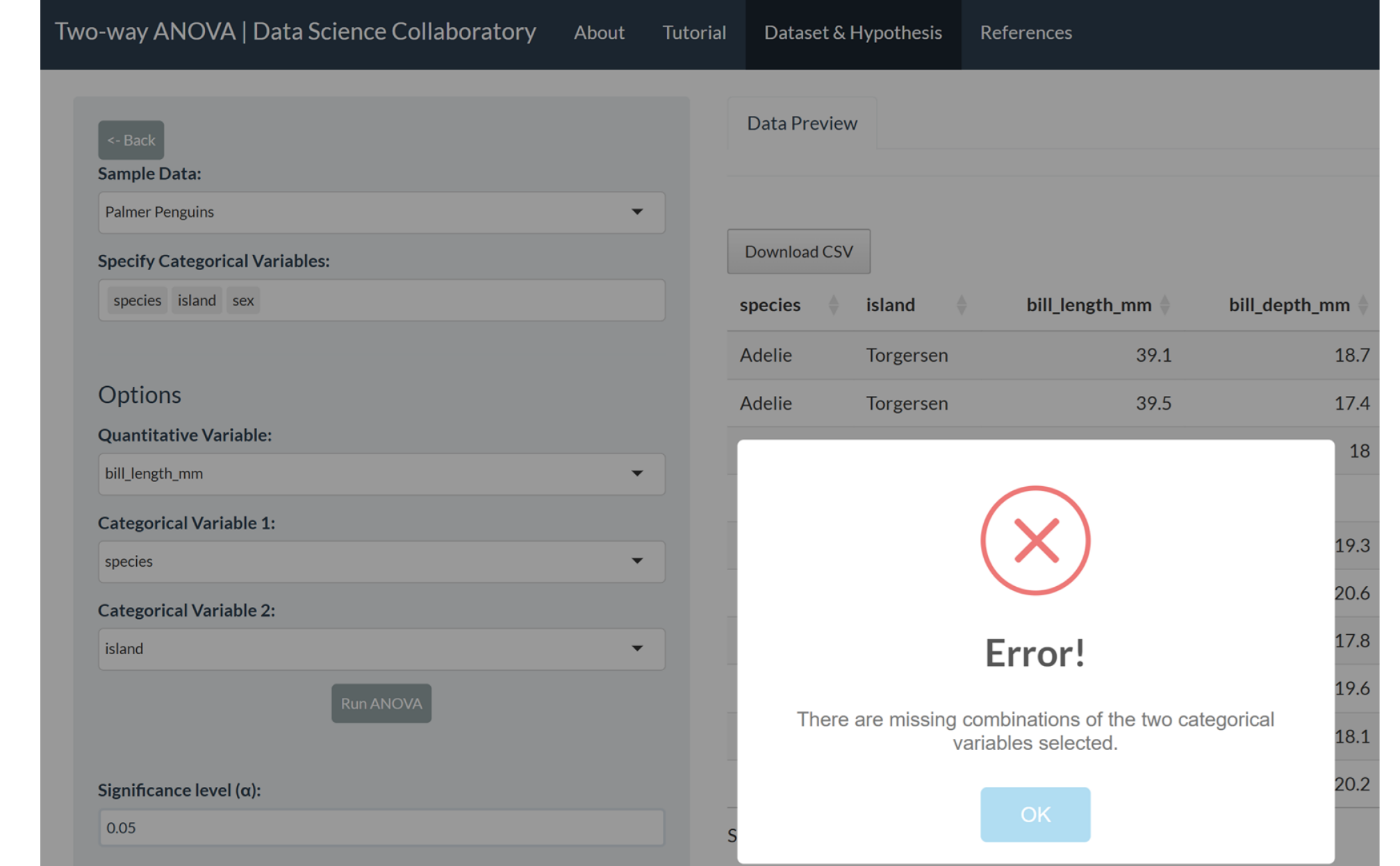
We've added cumulative statistics and noodle plots to our Resampling app, illustrating the Law of Large Numbers: sample statistics converge to population statistics as the sample size increases.

TWO-WAY ANOVA



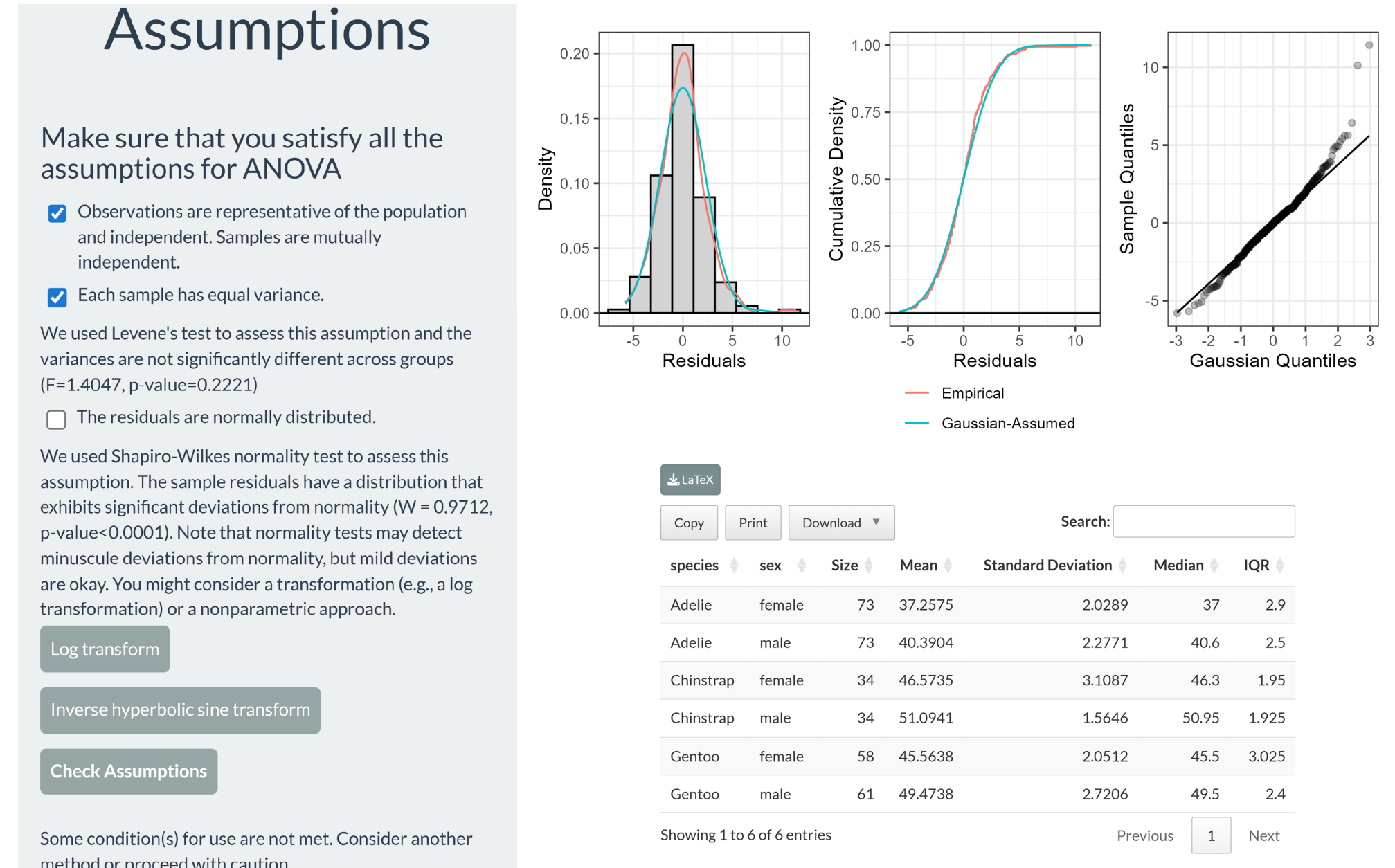
The new Two-Way ANOVA app analyzes the effect of two factors and their interaction on the mean of a continuous variable.

APP REFINEMENTS



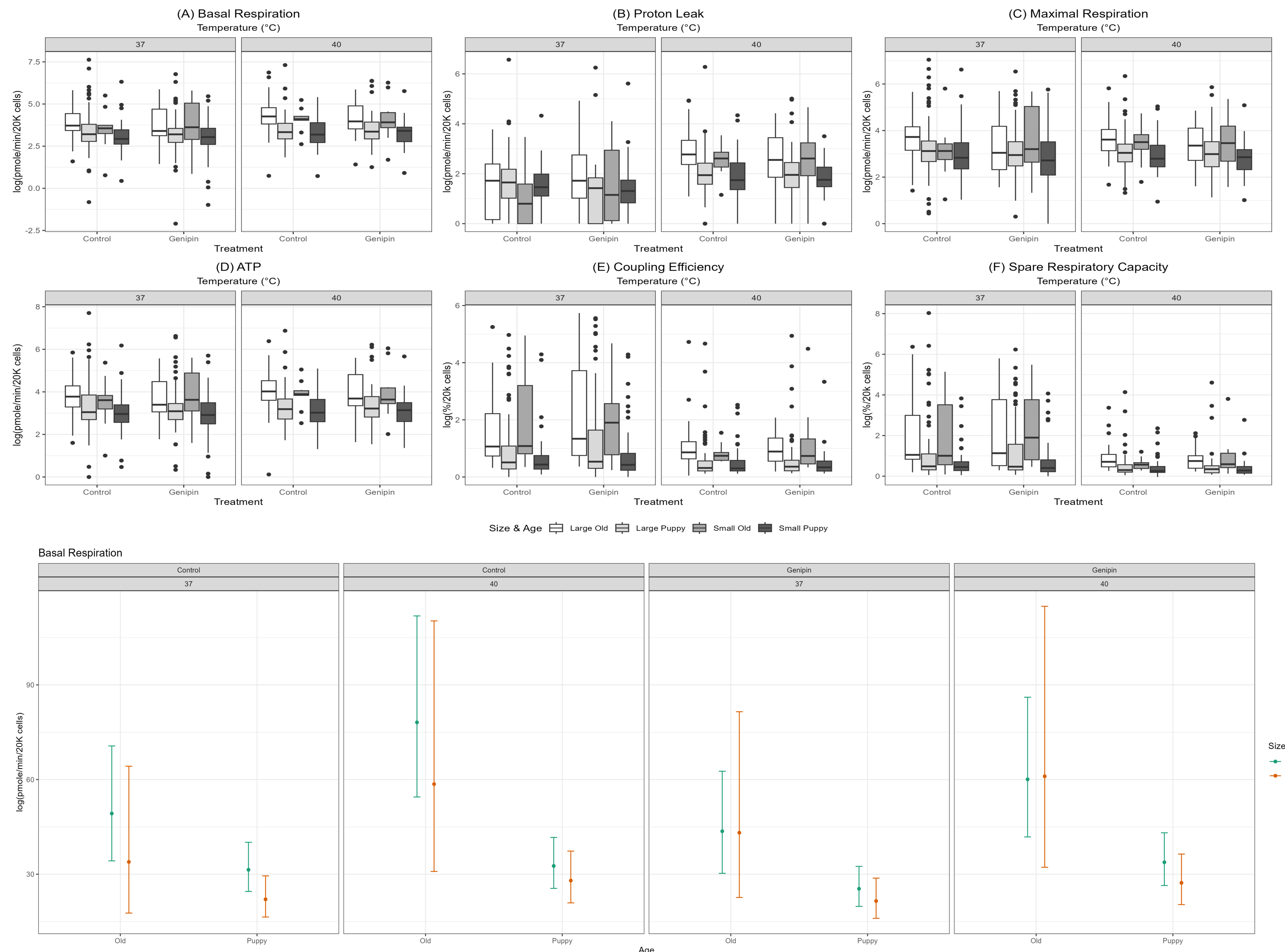
Our new “compute” buttons and app refinements enabled us to substantially improve app performance, including snappier transitions across tabs and more helpful error messages.

CHECKING ASSUMPTIONS



We've made the assumption checking tab clearer and more educational. Users now have the option to transform non-Gaussian data.

MIXED EFFECTS REGRESSION PROJECT W/ PROF. JIMENEZ



- We developed an alpha version of our Mixed Effects Regression app alongside a project with Prof. Jimenez, which uses these models.
- This is our first and currently only app that incorporates random effects or intercepts, which are often necessary for experimental data.
- Preliminary results suggest mitochondrial function varies across dogs depending on size and age.
- Genipin treatment inhibits part of mitochondrial function and is often moderated by temperature.